

LANGUAGE: C PROGRAMMING LEVEL:BASIC-INTERMEDIATE

Day	Topic	Subtopics	Interview Coding Problems (LeetCode / HackerRank)	End of Module Use Case
Day 1	C Fundamentals	Introduction to C, Compilation Process, Identifiers, Keywords, Variables, Constants, Data Types, Type Casting	Two Sum, Palindrome Number, Hello World in C	Student Grade Calculator
Day 2	Operators & Expressions	Arithmetic, Relational, Logical, Bitwise, Assignment, Increment/Decrement, Operator Precedence	Reverse Integer, Number of 1 Bits, Bitwise Operators	Scientific Calculator
Day 3	Conditional Statements & Loops	if-else, switch-case, for, while, do-while, Nested Loops, break & continue	Fizz Buzz, Valid Perfect Square, Pattern Printing	ATM Menu Driven System
Day 4	Functions & Recursion	Function Declaration, Call by Value/Reference, Recursion, Storage Classes	Fibonacci Number, Power of Two, Recursive Digit Sum	Electricity Bill Generator
Day 5	Arrays	1D/2D Arrays, Searching, Sorting Basics, Matrix Operations	Maximum Subarray, Best Time to Buy & Sell Stock	Student Marks Analysis System
Day 6	Strings	String Functions, String Manipulation, Character Arrays, Command Line Arguments	Valid Anagram, Longest Common Prefix	Password Validation System
Day 7	Pointers	Pointer Basics, Pointer Arithmetic, Arrays with Pointers, Functions with Pointers	Move Zeroes, Merge Strings Alternately	Dynamic Employee Record Access
Day 8	Advanced Pointers	Double Pointers, Function Pointers, Dangling/Wild/Null Pointers	Remove Duplicates from Sorted Array	Menu Driven Calculator using Function Pointers
Day 9	Structures & Memory	Structures, Unions, Enumeration, typedef, Structure Padding & Alignment	Richest Customer Wealth, Find Pivot Index	Employee Management System
Day 10	Dynamic Memory & Files	Stack vs Heap, malloc/calloc/realloc/free, File Handling, Read/Write Structures	Contains Duplicate, Remove Element	Inventory Management using Files
Day 11	Error Handling & Debugging	perror(), errno, Assertions, goto for cleanup, gdb basics, Segmentation Faults	Missing Number, Climbing Stairs	Banking System with Error Logs
Day 12	Advanced Interview Problem Solving	Time Complexity, Space Complexity, Advanced Arrays, Strings, Hashing, Sliding Window, Sorting & Searching, Pattern-Based Coding Questions, Output Prediction	LeetCode: Two Sum, Maximum Subarray, Valid Parentheses, Merge Sorted Array, Best Time to Buy & Sell Stock, Majority Element HackerRank: Arrays DS, Left Rotation, Sparse Arrays, Counting Valleys, Balanced Brackets	Campus Interview Coding Round Simulation

Day	Topic	Subtopics	Interview Coding Problems (LeetCode / HackerRank)	End of Module Use Case
Day 13	Advanced Coding + RDBMS Concepts	First Half – Coding Practice: Mock Coding Round, Debugging Problems, Logical Reasoning through Coding, Interview MCQs, Optimization Techniques. Second Half – RDBMS Concepts: DBMS vs RDBMS, ER Model, Primary Key, Foreign Key, Normalization, SQL Basics (SELECT, INSERT, UPDATE, DELETE), Joins, Aggregate Functions	LeetCode: Binary Search, Rotate Array, Contains Duplicate, Valid Anagram, Climbing Stairs HackerRank: SQL Basic Select, Weather Observation Station, Dynamic Array, Problem Solving (Easy-Medium)	Student Database Management System
Day 14	Cloud, Cloud Basics, GitHub, Docker, Kubernetes & CI/CD	Overview of Cloud Computing, Types of Cloud Services (IaaS, PaaS, SaaS), Cloud Deployment Models, Virtualization, Git & GitHub Basics, clone/push/pull/merge conflicts, Docker Basics, Docker Commands, Docker Images, Docker Networking & Volumes, Docker Compose, Kubernetes Architecture & Components, K8S Commands, Deploying Applications on K8S, K8S Networking & Service Discovery, Introduction to DevOps, CI/CD Pipelines, Infrastructure as Code (IaC), GitHub Actions/Jenkins Basics	-	End-to-End DevOps Deployment Pipeline using GitHub + Docker + Kubernetes

Key Interview Preparation Areas

Area	Important Concepts
C Programming Core	Arrays, Strings, Pointers, Structures
Problem Solving	Searching, Sorting, Pattern Logic
Memory Concepts	Heap, Stack, Memory Leak
Debugging	Segmentation Fault, Null Pointer
Git & GitHub	clone, commit, push, pull request, merge conflict
Cloud Basics	AWS/Azure/GCP concepts, Deployment Models
Docker	Dockerfile, Images, Containers, Volumes
Kubernetes	Pods, Services, Deployments
DevOps	CI/CD Pipelines, GitHub Actions, IaC Basics

Recommended Practice Platforms

[LeetCode](#) , [HackerRank C Programming](#), [GeeksforGeeks C Programming](#), [Docker Documentation](#), [Kubernetes Documentation](#), [GitHub Docs](#)